

Iuliia Thorbecke (Nigmatulina)

julia.nigmatic@gmail.com



About me

Machine Learning and Speech Recognition specialist with a PhD in ASR and ~5 years of applied research experience. Expertise in contextual biasing, streaming ASR, and real-world deployment in high-stakes environments (air traffic communication, call centers). Background in low-resource speech systems, with hands-on experience in PyTorch, Kaldi, and modern end-to-end ASR frameworks.

Personal

Iuliia Thorbecke
residence: Switzerland
(spouse of a Swiss citizen)
03.05.1990

Skills

Machine Learning & ASR:

- End-to-end (Transducers, CTC, attention) and hybrid ASR
- Streaming ASR, contextual biasing, domain adaptation
- Language modelling, decoding (beam search, FST)

Frameworks:

- PyTorch, TensorFlow, Hugging-Face, SLAM-LLM
- Kaldi, K2/Icefall, SpeechBrain
- PRAAT, Audacity, PROSOGRAM

Programming:

- Python; Bash (advanced)
- C++ (practical familiarity through Kaldi); JavaScript; R (basic)
- Linux/GPU environments

Data & Systems:

- SQL, Elasticsearch, large-scale data processing

Languages

- English-C1 • German-B2
- French-B2 • Russian-native

Iuliia-Thorbecke

Google-Scholar

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WORKING EXPERIENCE

2020–2025

Research Assistant (full-time)

IDIAP RESEARCH INSTITUTE · Martigny, Switzerland 📍

- Developed streaming ASR systems for air-traffic communication achieving WER below 5% in noisy, safety-critical environments (collaboration with *The German Aerospace Center (DLR)*)
- Designed and implemented contextual biasing methods to improve recognition of rare and domain-specific vocabulary in real-time ASR decoding with dynamic context adaptation (FST, beam search, language modelling)
- Contributed to production-oriented ASR systems for call center applications, focusing on scalability, domain adaptation, and robustness to real-world variability (collaboration with *Uniphore*)
- Published and co-published 25 scientific articles in the top-tier ASR and NLP conferences (ICASSP, INTERSPEECH, EMNLP)
- Best paper award (ICASSP Workshop 2025) for the paper "*Performance evaluation of slam-asr: The good, the bad, the ugly, and the way forward*"

2018

Intern (full-time)

SPINNINGBYTES AG · Winterthur, Switzerland 📍

Worked on different machine learning tasks, such as *text classification* and *sentiment analysis*, using a large collection of text data (tweets) organised with the Elasticsearch engine

2017–2020

Student Assistant

INSTITUTE OF COMPUTATIONAL LINGUISTICS, UNIVERSITY OF ZURICH · Zurich, Switzerland 📍

- Engineered a rule-based pipeline to automate *stance detection* across newspaper datasets
- Speech segmentation; phonetic transcription; speech experiments; iVector extraction
- Built dialectal ASR systems for Swiss German speech processing
- Designed and conducted an introductory course to Python for phoneticians

2011–2016

Engineer (full-time)

SPEECH MODELING LABORATORY, ST. PETERSBURG STATE UNIVERSITY · St. Petersburg, Russia 📍

- Creation of a spontaneous speech corpus
- Analysed acoustic realisations of spontaneous speech
- Designed and conducted speech experiments; phonetic transcription of speech
- Annotated and analysed intonation contour of spontaneous speech

DEGREES

2021–2025

PhD in Speech Technologies

THESIS: "CONTEXTUALISATION OF AUTOMATIC SPEECH RECOGNITION AND RELATED APPLICATIONS"
UNIVERSITY OF ZURICH · Zurich, Switzerland 🏛️

2016–2020

Master's degree in Computational Linguistics and Speech Technologies

THESIS: "ACOUSTIC MODELLING FOR SWISS GERMAN ASR"
UNIVERSITY OF ZURICH · Zurich, Switzerland 🏛️

2007–2013

Master's degree in Psycholinguistics and General Linguistics

THESIS: "SOUND CONTRACTIONS IN RUSSIAN SPONTANEOUS AND PREPARED SPEECH" (FOCUS ON SPEECH PERCEPTION ALGORITHMS)
ST. PETERSBURG STATE UNIVERSITY · St. Petersburg, Russia 🏛️

TEACHING EXPERIENCE

2018–2020	Tutor (Institute of Computational Linguistics) UNIVERSITY OF ZURICH · Zurich, Switzerland 📍 "Language Technology and Web Applications" (HS18); "Speech Science and Speech Signal Processing" (FS19); "Machine Learning in Computational Linguistics" (HS19); "Text Analytics in the Digital Humanities" (FS20)
2014–2016	Lecturer ST. PETERSBURG STATE UNIVERSITY OF AEROSPACE INSTRUMENTATION (THE FACULTY OF ART) · St. Petersburg, Russia 📍 "Introduction to General Phonetics"; "Methods of speech analysis"

PROJECTS

- **Air-Traffic ASR** (DLR) "STARFISH": Safety and Speech Recognition with Artificial Intelligence in the Use of Air Traffic Control. IDIAP, Switzerland
- **Call Center ASR** (Uniphore): Domain adaptation and contextual biasing. IDIAP, Switzerland
- **Swiss German ASR**: Acoustic modelling for low-resource dialects. University of Zurich, Switzerland
- **Recognition of ambiguous fragments of speech signal** (evidence from spontaneous Russian). St. Petersburg State University, Russia
- **Role of pitch** in the preservation of unity of "broken" discourse units in spontaneous speech. St. Petersburg State University, Russia
- **Second language perceptual experience** and first language sound identification (part-time collaboration). Purdue University, Purdue, USA

SELECTED PUBLICATIONS (FULL LIST OF PUBLICATIONS AT GOOGLE-SCHOLAR)

Thorbecke, I., Villatoro-Tello, E., Pablo Zuluaga, J., Kumar, S., Burdisso, S., Rangappa, P., Carolis, A., Madikeri, S., Motlicek, P., Pandia, K., Hacıoğlu, K., Stolcke, A. (2025). *Unifying Global and Near-Context Biasing in a Single Trie Pass*. International Conference on Text, Speech, and Dialogue (**TSD** 2025). Pp. 170-181.

Kumar, S., Thorbecke, I., Burdisso, S., Villatoro-Tello, E., KE, M., Hacıoğlu, K., Rangappa, P., Motlicek, P., Ganapathiraju, A., Stolcke, A. (2024). *Performance evaluation of slam-asr: The good, the bad, the ugly, and the way forward*. In 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (**ICASSPW**) (pp. 1-5). IEEE. **[Best paper award]**

Thorbecke, I., Zuluaga-Gomez, J., Villatoro-Tello, E., Kumar, S., Rangappa, P., Burdisso, S., Motlicek, P., Pandia, K., Ganapathiraju, A. (2024). *Fast Streaming Transducer ASR Prototyping via Knowledge Distillation with Whisper*. Findings-**EMNLP** 2024.

Nigmatulina, I., Madikeri S., Villatoro-Tello E., Motlicek P., Zuluaga-Gomez J., Pandia K., Ganapathiraju A. (2023). *Implementing contextual biasing in GPU decoder for online ASR*. Proc. **Inter-speech** 2023.

Nigmatulina, I., Zuluaga-Gomez, J., Prasad, A., Sarfjoo, S. S., & Motlicek, P. (2022). *A two-step approach to leverage contextual data: speech recognition in air-traffic communications*. In ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**).

Nigmatulina, I., Kew, T., and Samardžić, T. (2020). *ASR for Non-standardised Languages with Dialectal Variation: the case of Swiss German*. **VarDial** workshop, 2020.

Kew, T., Nigmatulina, I., Nagele, L., and Samardžić, T. (2020). *UZH TILT: A Kaldi recipe for Swiss German Speech to Standard German Text*. **SwissText** conference, 2020.

✉ julia.nigmatic@gmail.com 📍 Unterägeri, Switzerland 📞 0041/79 9162977